

Problem #1 (15-points)

$$D_{11} = D_{22} = D_{33} = D_{44} = GMR_S = 0.0355 \text{ ft}$$

$$D_{12} = D_{21} = \frac{6}{12} = 0.5 \text{ ft}, \quad D_{34} = D_{43} = \frac{3}{12} = 0.25 \text{ ft}$$

$$D_{13} = D_{31} = D_{23} = D_{32} = \sqrt{\left(\frac{3}{12}\right)^2 + \left(\frac{3}{12}\right)^2} = 0.35355 \text{ ft}$$

$$D_{14} = D_{41} = D_{24} = D_{42} = \sqrt{\left(\frac{3}{12}\right)^2 + \left(\frac{6}{12}\right)^2} = 0.55902 \text{ ft}$$

$$GMR = \left[(0.0355)^4 (0.5)^2 (0.25)^2 (0.35355)^4 (0.55902)^4 \right]^{1/16} = 0.223173 \text{ ft}$$

Problem #2 (25-points)

$$\ell = 32 \text{ mi}, \quad \bar{\gamma} = 0.0039 \angle 73.26^\circ / \text{mi}, \quad \bar{Z}_c = 528.57 \angle -9.87^\circ \Omega$$

$$\bar{\gamma} \cdot \bar{Z}_c = \sqrt{\bar{\gamma} \cdot \bar{y}} \cdot \sqrt{\bar{\gamma} / \bar{y}} = \sqrt{\bar{\gamma}^2} = \bar{\gamma} = 2.061423 \angle 63.390^\circ \Omega / \text{mi}$$

$$\bar{\gamma} / \bar{Z}_c = \sqrt{\bar{\gamma} \cdot \bar{y}} / \sqrt{\bar{\gamma} / \bar{y}} = \sqrt{\bar{y}^2} = \bar{y} = 7.378398 \times 10^{-6} \angle 83.130^\circ \text{ S} / \text{mi}$$

Medium Length Model ABCD Parameters

$$\bar{Z}_{TL} = \bar{\gamma} \ell = 65.965536 \angle 63.390^\circ \Omega, \quad \bar{Y}_{TL} = \frac{1}{2} \bar{y} \ell = 1.180544 \times 10^{-4} \angle 83.130^\circ \text{ S}$$

$$\bar{A} = \bar{D} = 1 + \bar{Z}_{TL} \bar{Y}_{TL} = 0.993514 \angle 0.248^\circ$$

$$\bar{B} = \bar{Z}_{TL} = 65.965536 \angle 63.390^\circ$$

$$\bar{C} = \bar{Y}_{TL} (2 + \bar{Z}_{TL} \bar{Y}_{TL}) = 2.353425 \times 10^{-4} \angle 83.253^\circ$$

Long Length Model ABCD Parameters

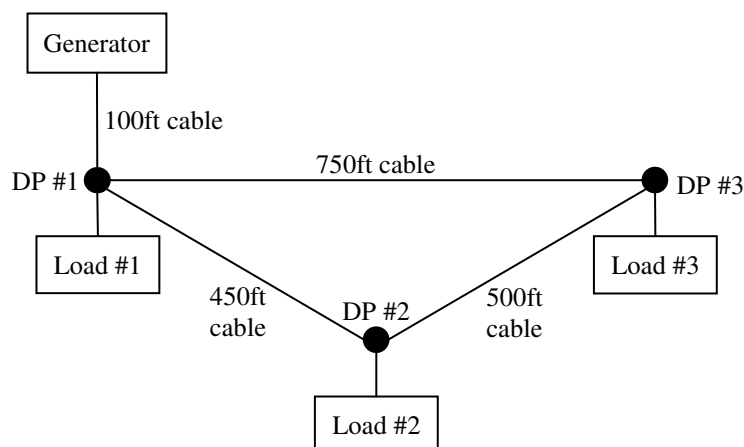
$$\bar{A} = \bar{D} = \cosh(\bar{\gamma} \ell) = 0.993518 \angle 0.247^\circ$$

$$\bar{B} = \bar{Z}_c \sinh(\bar{\gamma} \ell) = 65.822832 \angle 63.472^\circ$$

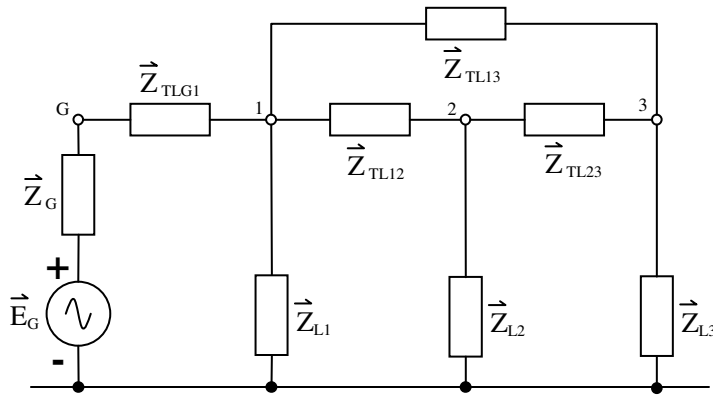
$$\bar{C} = \bar{Z}_c^{-1} \sinh(\bar{\gamma} \ell) = 2.355980 \times 10^{-4} \angle 83.212^\circ$$

Problem #3 (25-points)

Step 1: Visualize



Step 2: Equivalent Circuit



Step 3: Circuit Parameters

Generator:

$$S_{G3\phi rated} = 500 \text{ KVA}, \quad V_{GLL rated} = 480 \text{ V}, \quad X_G = 21\%, \quad X/R = 15, \quad V_{GLLNL} = 515 \text{ V}$$

$$\bar{E}_G = \left(\frac{V_{GLLNL}}{\sqrt{3}} \right) \angle 0^\circ \text{ V} = 297.3354 \angle 0^\circ \text{ V}$$

$$R_G = X_G / (X/R) = 1.4\%, \quad \bar{Z}_G = \left(\frac{R_G + jX_G}{100} \right) \left(\frac{0.480^2}{0.500} \right) = 0.096983 \angle 86.186^\circ \Omega$$

Loads:

$$\bar{Z}_{L1} = \bar{Z}_{L2} = \bar{Z}_{L3} = \frac{1}{3} \bar{Z}_\Delta = \frac{1}{3} (12 \angle 37^\circ \Omega) = 4 \angle 37^\circ \Omega$$

Power Cables:

$$R = \frac{R_s}{n} = \frac{1.518 \Omega / mi}{1} = 1.518 \Omega / mi$$

$$GMR = e^{-0.25} r_s = e^{-0.25} (0.127 \text{ in}) = 0.09891 \text{ in}, \quad GMD_{eq} = \sqrt[3]{D_{AB} D_{BC} D_{CA}} = \sqrt[3]{(1 \text{ in})(1 \text{ in})(2 \text{ in})} = 1.2599 \text{ in}$$

$$L = 1.03 \cdot k_{L3} \cdot \ln(GMD_{eq} / GMR) = 8.436039 \times 10^{-4} \text{ H} / mi$$

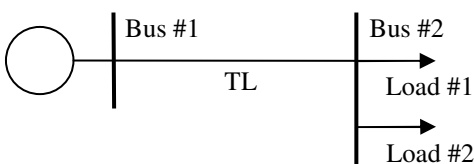
$$\bar{Z} = R + j\omega L = 1.550957 \angle 11.833^\circ \Omega / mi$$

$$\bar{Z}_{TLG1} = \left(\frac{100 \text{ ft}}{5280 \text{ ft} / mi} \right) \bar{Z} = 0.02937 \angle 11.833^\circ \Omega, \quad \bar{Z}_{TL12} = \left(\frac{450 \text{ ft}}{5280 \text{ ft} / mi} \right) \bar{Z} = 0.13218 \angle 11.833^\circ \Omega$$

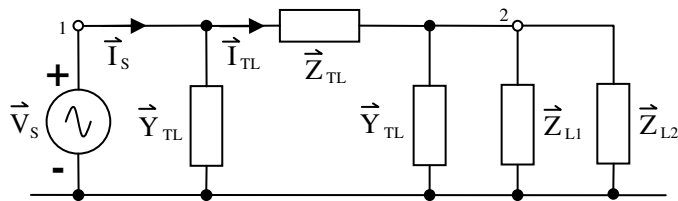
$$\bar{Z}_{TL23} = \left(\frac{500 \text{ ft}}{5280 \text{ ft} / mi} \right) \bar{Z} = 0.14687 \angle 11.833^\circ \Omega, \quad \bar{Z}_{TL13} = \left(\frac{750 \text{ ft}}{5280 \text{ ft} / mi} \right) \bar{Z} = 0.22031 \angle 11.833^\circ \Omega$$

Problem #4 (35-points)

Step 1 – Visualize



Step 2 – Equivalent Circuit



Step 3 – Circuit Parameters

$$\vec{V}_s = 5000 \angle 0^\circ \text{ V}, \quad \vec{Z}_{L1} = 300 \angle -35^\circ \Omega, \quad \vec{Z}_{L2} = 200 \angle 45^\circ \Omega$$

$$\ell = 2 \text{ mi}, \quad R = \frac{R_{xx}}{n_x} + \frac{R_{yy}}{n_y} = \frac{0.1342}{1} + \frac{0.1342}{1} = 0.2684 \Omega / \text{mi}$$

$$GMR_x = GMR_y = GMR_s = 0.0355 \text{ ft}, \quad mGMR_x = mGMR_y = r_s = 0.0417 \text{ ft}, \quad GMD_{xy} = 3 \text{ ft}$$

$$L = k_{L1} \cdot \ln \left(\frac{GMD_{xy}}{\sqrt{GMR_x GMR_y}} \right) = 2.85615743 \times 10^{-3} \text{ H / mi}, \quad C = k_{C1} \cdot \ln \left(\frac{GMD_{xy}}{\sqrt{mGMR_x mGMR_y}} \right) = 1.046943 \times 10^{-8} \text{ F / mi}$$

$$\vec{\gamma} = R + j120\pi L = 1.109694 \angle 76.003^\circ \Omega / \text{mi}, \quad \vec{y} = G + j\omega C = 3.946884 \times 10^{-6} \angle 90^\circ \text{ S / mi}$$

$$\vec{Z}_{TL} = \vec{\gamma} \ell = 2.219388 \angle 76.003^\circ \Omega, \quad \vec{Y}_{TL} = \frac{1}{2} \vec{y} \ell = 3.946884 \times 10^{-6} \angle 90^\circ \text{ S}$$

Step 4: Circuit Analysis

$$\vec{Z}_{eq2} = (\vec{Y}_{TL} + \vec{Z}_{L1}^{-1} + \vec{Z}_{L2}^{-1})^{-1} = 154.512114 \angle 14.493^\circ \Omega$$

$$\vec{Z}_{eq1} = \left[\vec{Y}_{TL} + \left[\vec{Z}_{TL} + \vec{Z}_{eq2}^{-1} \right]^{-1} \right]^{-1} = 155.608033 \angle 15.177^\circ \Omega$$

$$\vec{I}_s = \vec{V}_s / \vec{Z}_{eq1} = 32.132017 \angle -15.177^\circ \text{ VA}, \quad \vec{S}_s = \vec{V}_s \vec{I}_s^* = 160,660.0859 \angle 15.177^\circ \text{ VA},$$

$$\vec{I}_{TL} = \vec{I}_s - \vec{Y}_{TL} \vec{V}_s = 32.137189 \angle -15.211^\circ \text{ A}$$

$$\vec{V}_2 = \vec{V}_s - \vec{Z}_{TL} \vec{I}_{TL} = 4965.585068 \angle -0.718^\circ \text{ V}$$

Step 5: Final Results

Real and reactive power supplied by the source:

$$P_s = \text{Re}\{\vec{S}_s\} = 155.0564 \text{ KW}, \quad Q_s = \text{Im}\{\vec{S}_s\} = 42.0616 \text{ KVAR}$$

Voltage at the loads:

$$V_2 = |\vec{V}_2| = 4.9656 \text{ KV}$$